



User Guide v2.0 (May 2026)

Drug Repositioning and Synthetic Patient Treatment Decision Support System for Childhood Acute Leukemia

TÜBİTAK 1001 · Project No: 123E383 · Süleyman Demirel University · 2026
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(Generated through Human-AI collaboration. AI was used to accelerate content generation and human checked – confirmed the final format)

1. Getting Started

STING DSS is a web-based decision support platform for childhood acute lymphoblastic leukaemia (ALL). It brings together five analytical layers — Bi-LSTM drug repositioning, ODE-based PK/PD simulation, genetic algorithm dose optimisation, GNN digital twin prediction, and CTGAN synthetic patient generation — into a single, step-by-step workflow that runs entirely in your browser. No installation is needed on your computer.

The six tabs in the left sidebar form a pipeline: each tab's output feeds the next one. You can always jump back to adjust parameters, but the first time through, work left to right.

1.1 Signing In

Open the system URL in any modern browser. The login card appears in the centre of the screen. Two small controls in the top corners of the card are worth noticing before you type anything:

- Language selector (top-left of the card) — the system detects your browser language automatically. If it has picked the wrong one, click 'TR' or 'EN' to switch. The change takes effect immediately on the current screen.
- Theme toggle (top-right, ☀/🌙 icon) — switches between light and dark mode. Both are fully supported throughout the application; choose whichever is more comfortable.

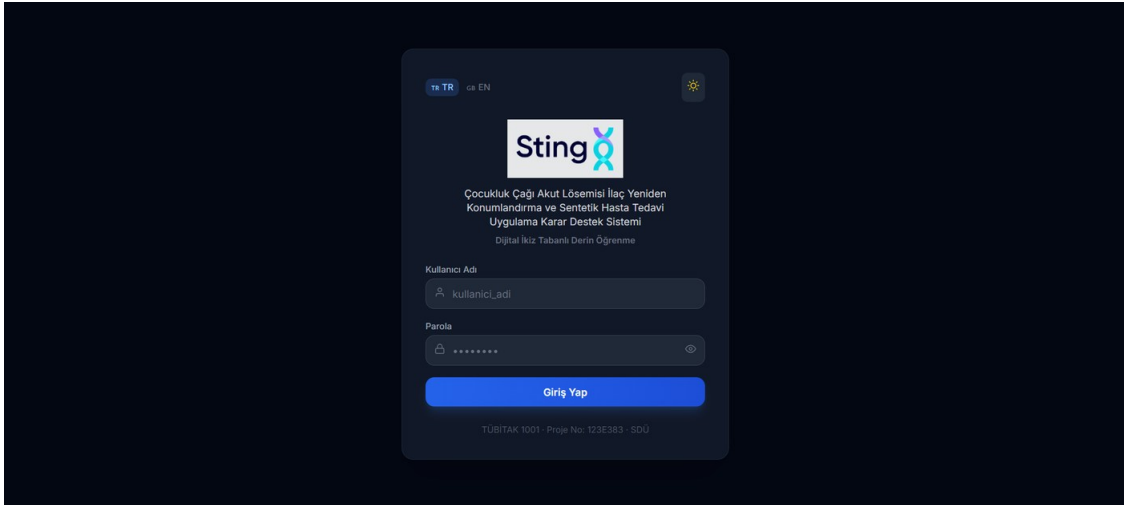


Figure 1. Login screen — language selector top-left, theme toggle top-right. The subtitle confirms this is the TÜBİTAK 1001 / 123E383 system.

Type your username into the first field and your password into the second, then click 'Sign In'. The eye icon at the right of the password field reveals the characters if you want to check for typos. If everything is correct, the application loads Tab 1 within a few seconds.



Note

After 10 consecutive failed login attempts the account is temporarily locked. If this happens, ask your administrator to reset the password from the Admin Panel.

1.2 Getting Around the Interface

Once logged in, the screen is divided into three permanent zones that stay visible across all tabs:

Zone	Where it is	What it does
Pipeline sidebar	Left edge	Lists all six workflow steps by name and number. The currently active tab is highlighted. Any tab that has been completed shows a green ✓ next to its number. Tabs that are not yet unlocked appear dimmed.
Top bar	Top of page	Shows the session name field (click 'New Project' to rename it), Save / Open / New buttons, the sequential tab numbers 1–6 for quick jumping, and the language switcher.
User zone	Bottom-left of sidebar	Displays your username and role. The moon/sun icon switches the theme. The arrow icon logs you out. The 'Survey' and 'About' buttons also live here.

Tip Use 'Save' in the top bar at the end of each tab. Sessions persist on the server, so you can close the browser and resume later by clicking 'Open' and choosing your session name.

2. Tab 1 — Drug Repositioning

This is where the research journey begins. Tab 1 uses a Bi-LSTM deep learning model — trained on more than 314,000 ligand–protein pairs drawn from DrugBank, ChEMBL, PubChem, and KIBA — to score which existing drugs might bind relevant ALL targets. The tab is split into three sub-tabs: Prediction, Molecular Docking, and Model Training. For most sessions you will only need the first two.

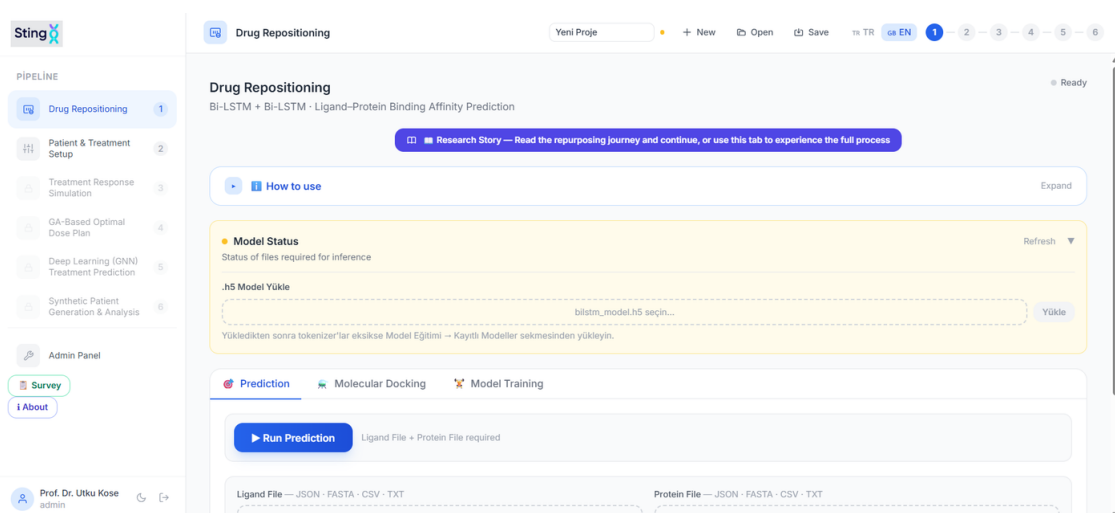


Figure 2. Tab 1 on first load. The yellow Model Status panel is at the top; the blue Research Story banner sits below it; the three sub-tab buttons (Prediction · Molecular Docking · Model Training) are in the middle of the page.

2.1 Checking the Model Status Panel

The yellow 'Model Status' panel near the top of the page is the first thing to check every time you open Tab 1. It reports whether the Bi-LSTM model file and its tokenizers are loaded and ready. The indicator dot on the right of the panel header tells the story at a glance:

- Green dot — everything is loaded. You can go straight to the Prediction sub-tab.
- Yellow dot — the model file is present but tokenizer files are missing. Click 'Refresh' on the panel to re-check, or navigate to Model Training → Saved Models to upload the tokenizer pickle files.
- Red or missing — no model detected. Use the '.h5 Model Upload' drag-and-drop area inside the panel to select your model file. The upload completes in a few seconds.



Tip

The pre-trained model that ships with the system (bilstm_l2_bilstm_l2_hpo.h5) was selected after comparing five architectures. Unless you have trained your own model, leave the default in place and proceed to the Prediction sub-tab.

2.2 Reading the Research Story

Just below the model status panel you will see a wide purple banner that reads 'Research Story — Read the repurposing journey...' Click anywhere on this banner to open a six-slide walkthrough. The slides explain, step by step, how the project data was assembled, how the model was trained and validated, how molecular docking confirmed the top candidates, and why Copanlisib and Novobiocin were selected as the two repositioning drugs that appear throughout the rest of the pipeline.

The slides are not required reading to use the system, but they are the fastest way to understand the scientific rationale before examining any results. Use the 'Next →' and '← Previous' buttons to move between slides, and the × in the top-right corner to close the walkthrough when you are done.

2.3 Running a Prediction

Click the 'Prediction' sub-tab. Two upload zones appear side by side — one for ligands, one for proteins — and a 'Run Prediction' button sits below them. Work through the following steps:

1

Upload your ligand file

Drag the file onto the 'Ligand File' zone, or click the zone to open a file browser. Supported formats are JSON (a dictionary of compound name → SMILES string), FASTA, CSV, or plain TXT with one SMILES per line. The project's standard ligands.txt file uploads directly.

2

Upload your protein file

Do the same for the 'Protein File' zone. Supported formats are FASTA or a sequence JSON object. The project's proteins.txt file is compatible as-is.

3

Click 'Run Prediction'

The button activates as soon as both files are accepted (a small file name appears under each zone confirming the upload). A progress spinner appears; predictions typically complete in under a minute.

4

Examine the ranked results

A table appears listing candidate drugs sorted by predicted binding affinity score, highest first. Each row includes the compound name, predicted score, and links to ChEMBL and PubChem. Click the 'XAI Explanation' button on any row to open a panel that highlights which parts of the molecule the model focused on for that prediction.



Tip

Copanlisib and Novobiocin — the two drugs pre-loaded into Tabs 2 through 6 — came from a prediction run on the full project dataset. Your own run may surface different top candidates depending on your input files. Top candidates from this tab can then be docked in the Molecular Docking sub-tab to validate binding.

2.4 Molecular Docking (Optional Validation Step)

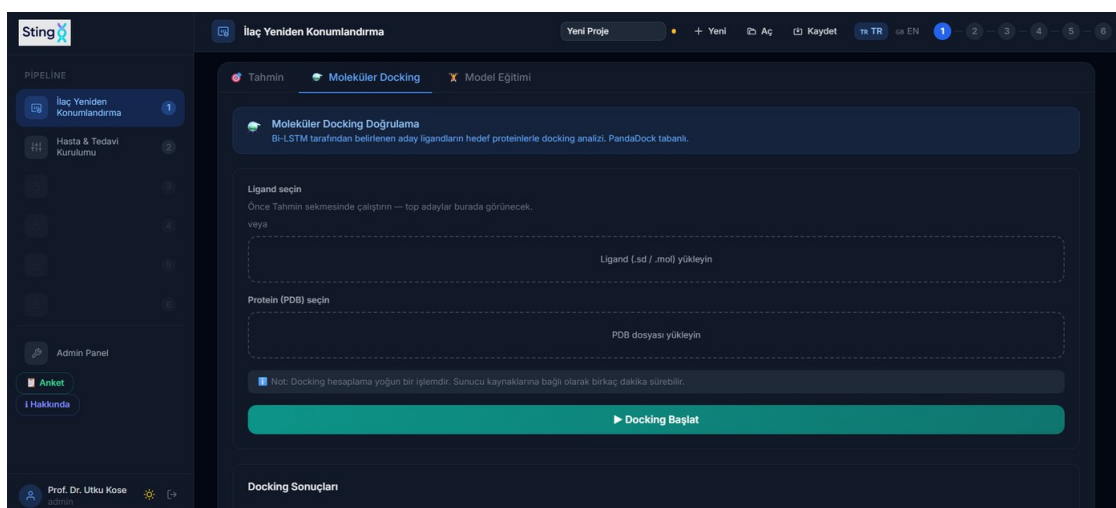


Figure 3. Molecular Docking sub-tab. The ligand upload zone (left) and protein upload zone (right) are both empty at the start. Once both files are uploaded, 'Start Docking' activates. The informational note below the zones reminds you that docking is compute-intensive and may take a few minutes.

Switch to the 'Molecular Docking' sub-tab if you want a structure-based confirmation of the binding predictions. This step is optional — you can proceed to Tab 2 without it. When you do run docking:

- Upload a ligand in 3D structure format (.sd or .mol — not a SMILES string this time, since docking requires 3D coordinates).
- Upload a receptor protein in PDB format.
- Click 'Start Docking'. The calculation runs on the server. A completion message appears when done, along with the docking score in kcal/mol and a visualisation of the binding pose.

For reference, in the project analysis Copanlisib docked against P02545 at -207.11 kcal/mol and Novobiocin at -168.43 kcal/mol — both indicating strong predicted binding.

3. Tab 2 — Patient and Treatment Setup

Tab 2 is where you define the patient and treatment parameters that the ODE simulation engine will use. It does not run any calculations itself — think of it as filling out a detailed intake form. Every field you enter here propagates directly into Tabs 3, 4, and 5, so take a moment to fill in values that reflect the patient or scenario you want to model.

When all required fields are complete, a blue button appears at the top of the tab: 'Go to Response Simulation (Tab 3)'. Click it when you are ready to run the simulation. The badge on the button updates in real time as you fill in fields, showing the active drug count, simulation length, patient weight, and ODE engine status.

3.1 Selecting a Protocol

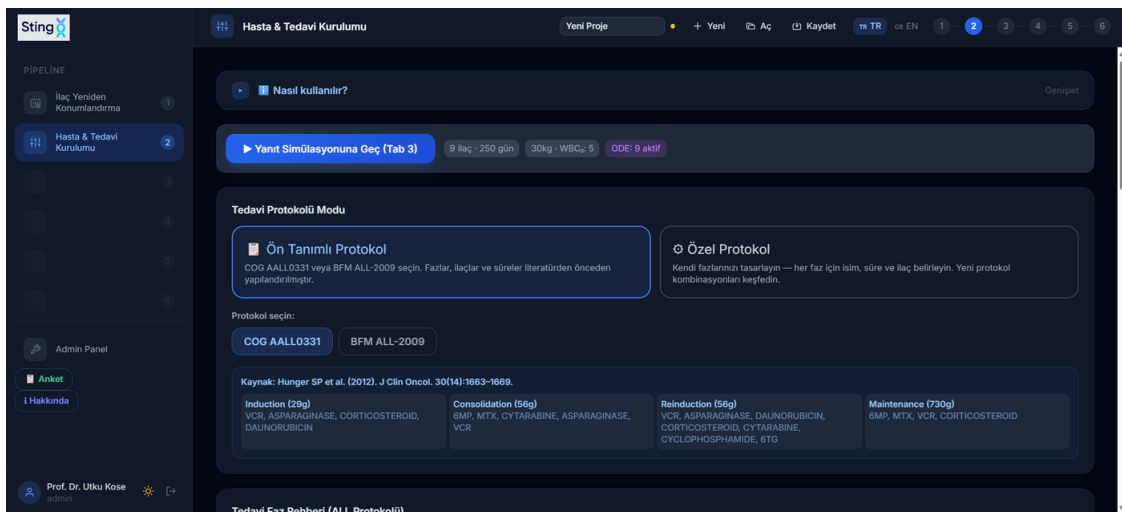


Figure 4. Protocol selection cards. The left card (Pre-defined Protocol) is selected and highlighted in blue. The COG AALL0331 and BFM ALL-2009 buttons appear below, and the selected protocol's phase structure is shown in the reference panel beneath them.

Two protocol modes are shown as side-by-side cards. Click the card that matches your scenario:

Mode	Best for	What happens when you select it
Pre-defined Protocol	Standard clinical simulations	Click either 'COG AALL0331' or 'BFM ALL-2009'. All phases, drug lists, durations, and default dose schedules load automatically. The phase summary panel below shows induction, consolidation, reinduction, and maintenance durations at a glance. You can still override individual doses afterward.
Custom Protocol	Research scenarios and novel combinations	Activates the Phase Designer, where you can create up to six phases from scratch. For each phase, set the name, the duration in days, and then click the drug pills to toggle individual drugs on or off. When a drug is toggled on, a scheduling row appears below it — leave it at 'default' unless you need a specific administration pattern. This is also where you activate Copanlisib (COP) and Novobiocin (NOV).

Figure 5. Custom Protocol Phase Designer. Phase 1 (129 days) has 6-MP and VCR highlighted in colour, meaning they are active. The scheduling row for 6-MP is open, showing 'default', 'daily oral', and 'weekly' options. COP and NOV pills appear at the right end of each drug row.



To include Copanlisib and Novobiocin in your simulation, switch to Custom Protocol and click the COP and NOV pills in the phase where you want them active. Their dedicated dose fields will then appear in the drug dose section below.

3.2 Patient Parameters

Scroll past the protocol section to the patient profile fields. Fill them from top to bottom — the system calculates BSA automatically from weight and height, so you never need to enter it manually.

Parameter	Unit	What to enter
Weight	kg	Body weight at treatment start. Used directly for BSA calculation and dose scaling.
Height	cm	Used with weight for BSA. Leave at 0 if unknown — doses will scale from weight alone.
Age	years	Used for NCI risk group assignment and age-related PK parameters.
Sex	M / F	Select from dropdown. Affects certain pharmacokinetic constants.
Initial WBC	$\times 10^9/L$	White blood cell count at diagnosis. Typical ALL presentation: 1–300.
Initial ANC	$\times 10^9/L$	Absolute neutrophil count at treatment start.
TPMT Genotype	normal / heterozygous / slow	Controls 6-mercaptopurine metabolism rate. Slow metabolisers require significant dose reduction.
Vitamin D	ng/mL	Influences immune modulation parameters in the ODE. Default: 30 ng/mL if unknown.

Diet Score	0–1	Nutritional status proxy. 1 = optimal; 0.75 is a reasonable default for most patients.
Exercise Score	0–1	Physical activity proxy. The sensitivity analysis in Tab 3 shows this has ~48% impact on WBC minimum — do not leave it at zero unless the patient is truly sedentary.
Infection	0 or 1	Set to 1 if an active infection is present at treatment start.



Note

Vitamin D level and Exercise Score look like minor lifestyle inputs, but Tab 3's sensitivity analysis consistently ranks them among the top three parameters influencing WBC minimum — on par with the patient's starting WBC itself. Enter realistic values rather than leaving them at zero.

3.3 Drug Doses

Below the patient fields, each active drug has its own dose input. The BSA-scaled equivalent appears beside the field and updates as you type. You can accept the protocol defaults (recommended for a first run) or enter custom values. Copanlisib (mg) and Novobiocin (mg/kg) appear at the bottom of the list only when those drugs are toggled on in the protocol.

When you are satisfied with all the parameters, click the blue 'Go to Response Simulation (Tab 3)' button at the top of the page.

4. Tab 3 — Treatment Response Simulation (ODE)

Tab 3 is the computational core of STING DSS. It feeds the patient and treatment parameters from Tab 2 into a system of ordinary differential equations that models how 10 drugs interact with leukaemic cell dynamics, healthy blood cell production, asparaginase activity, and toxicity indices over 250 days. The simulation uses a 48-dimensional state vector and typically solves in 3–8 seconds.

4.1 Starting the Simulation

When Tab 3 opens, the current drug doses and patient weight appear as coloured badges in the bar at the top of the page — for example, '6-MP 50 mg/g · MTX 20 mg/hf · VCR 1.5 mg/28g ...'. This is your last chance to confirm the inputs before committing to the run. If a dose looks wrong, click 'Tab 2 — Patient Setup' in the sidebar, correct it, and return.

Click 'Run Simulation'. A four-step progress panel counts off the stages:

- ODE initialising — assembling parameter vectors from your Tab 2 inputs
- Equations being solved — the RK45 adaptive solver integrating the state vector forward through 250 days
- Effects being computed — drug interaction terms, toxicity indices, and survival estimate calculated
- Graphs being generated — time-series data rendered into the interactive chart

When all four steps show green ticks and the timer in the top-right of the panel stops, the results appear. A green 'Completed!' banner summarises the headline metrics — for example, 'WBC min: 0.5968 · ANC min: 0.3446 · Target: 5.4% · PEG-ASP Asn min: 4.2 µmol/L'.

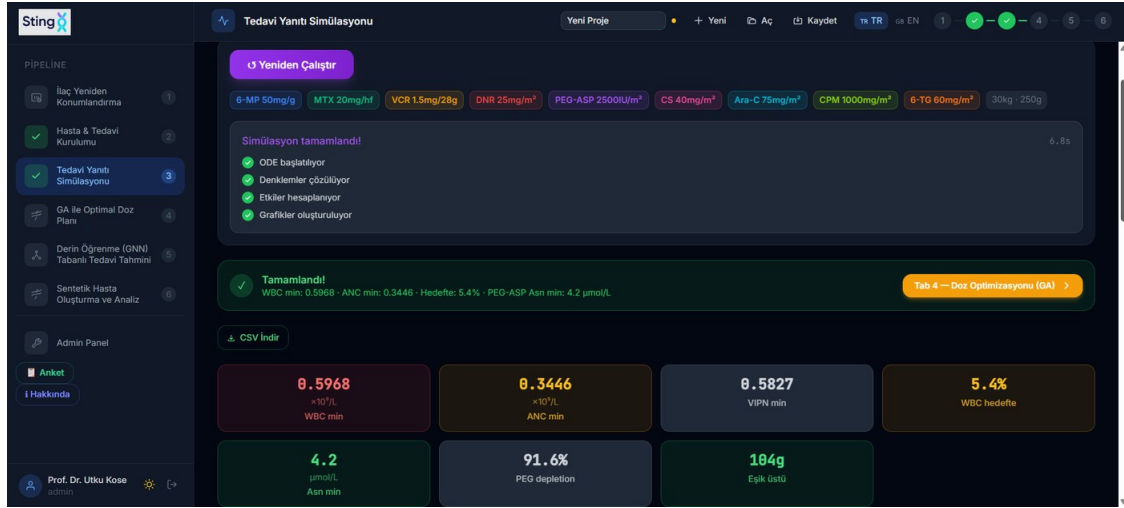


Figure 6. Tab 3 immediately after a successful simulation. The coloured metric cards fill the bottom half of the screen; the green completion banner with its quick-read summary sits above them. The 'Tab 4 — Dose Optimisation (GA)' button on the right of the banner is your one-click path to the next step.

4.2 Reading the Metric Cards

Six coloured cards appear below the completion banner. Each card shows a single value with its unit label and a contextual colour that signals whether the value is in a safe range. Here is what each card means and what to watch for:

Card	What it measures	Warning sign
WBC min	The lowest white blood cell count reached at any point in the 250-day simulation ($\times 10^9/L$). Reflects the depth of bone marrow suppression.	Below 0.5 indicates severe neutropenia territory; below 0.1 warrants urgent dose review.
ANC min	Minimum absolute neutrophil count — the most direct infection-risk indicator.	Below 0.5 is the febrile neutropenia threshold used in most clinical protocols.
VIPN min	Vincristine-Induced Peripheral Neuropathy index. Lower = more neuropathy predicted.	Below 0.78 is the safety threshold flagged in the system.
WBC target %	Percentage of simulation time WBC spends within the therapeutic target range.	Values below 5% suggest the protocol is not achieving adequate haematological control for this patient.

Asn min	Minimum PEG-Asparaginase activity in $\mu\text{mol/L}$. Depletion undermines anti-tumour efficacy.	Below 10 $\mu\text{mol/L}$ indicates asparaginase depletion; clinical supplementation may be warranted.
Cumulative DNR	Total daunorubicin dose accumulated across all four treatment phases (mg/m ²).	Above 300 mg/m ² crosses the internationally recognised cardiotoxicity limit.

4.3 Working with the Trajectory Chart



Figure 7. The 250-day trajectory chart. The solid blue line is WBC, the dashed green line is ANC, the dotted line is VIPN, and the rising orange curve in the maintenance phase is PEG-ASP activity recovering. The four coloured background regions (induction · consolidation · reinduction · maintenance) are labelled along the top.

The interactive chart below the metric cards is where you explore the full simulation trajectory. It is more than just a static plot — several controls let you dig into the data:

- Timeline slider (bottom of the chart) — drag the handle left or right to move through the 250 days one day at a time. The silhouette figure on the left side of the chart updates to reflect the patient's state on that day: the colour shifts from green through amber to red based on overall condition, and the WBC, ANC, and VIPN values beside the figure update numerically.
- Camera icon (top-right toolbar) — saves the current chart view as a PNG image. Use this to capture a specific time point or a zoomed-in region for a report.
- Magnifier icon — click then draw a rectangle on the chart to zoom into a specific time range. Double-click anywhere on the chart to zoom back out to the full 250-day view.
- Legend items (top of chart) — click any variable name in the legend to show or hide that line. If the chart looks cluttered, hide the drug effect curves and keep only WBC, ANC, and Lt visible.
- 'Download CSV' button (below the chart) — exports the entire 250-day time-series as a spreadsheet, with one row per day and columns for every tracked variable.



The vertical dashed lines on the chart mark phase transitions. The spike patterns you see around days 29, 85, and 141 are normal — they reflect scheduled dose pulses at phase changes, particularly the high-dose PEG-ASP injections.

4.4 Toxicity Analysis Panel

Scroll down past the chart to reach the 'Toxicity Analysis and Survival Estimate' section. This panel translates the simulation data into four clinically framed safety indicators:

- Febrile neutropenia days — the total number of days where ANC puts the patient in the febrile risk zone.
- Critical neutropenia days — days of profound neutropenia, shown alongside a 'threshold exceeded' flag if the count is high.
- DNR cumulative — the same value as the metric card, shown here alongside the 300 mg/m² limit line for easy visual comparison.
- VIPN min — shown here with a short plain-language description of the predicted neuropathy severity level.

The survival estimate shown in the top-right of this section (e.g., '100%') is a model-derived EFS probability based on the simulated trajectory. It is displayed for research exploration only and has not been validated against real clinical outcome data.

4.5 ODE Sensitivity Analysis

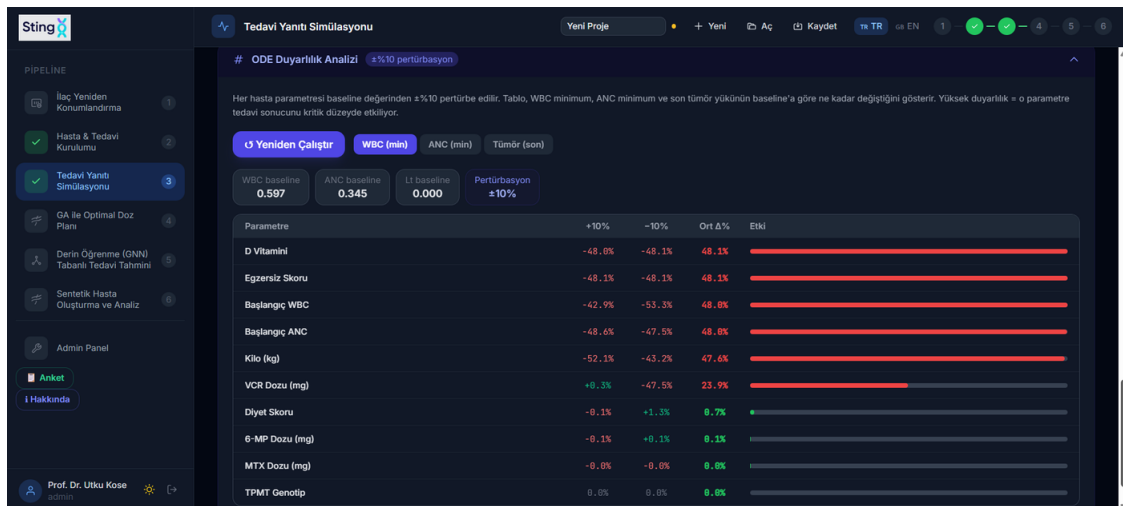


Figure 8. The Sensitivity Analysis table after clicking 'Re-run' with WBC min as the target. Each row is a patient parameter; the red horizontal bars show the average impact magnitude. Vitamin D and Exercise Score both reach 48.1% — higher than most drug doses.

At the very bottom of Tab 3, the 'ODE Sensitivity Analysis' section answers a critical question: among all the inputs you entered in Tab 2, which ones actually matter the most for the simulation outcome? It finds out by automatically re-running the simulation many times, each time nudging one parameter $\pm 10\%$ from its current value and recording how much the output changes.

To run it: choose a target metric using the three pill buttons at the top of the section (WBC min, ANC min, or Final tumour burden), then click 'Re-run'. The table fills in a few seconds. The three

numerical columns show the effect of a +10% increase, a -10% decrease, and the average absolute change (Avg $\Delta\%$). The bar on the right makes relative importance visible at a glance.

The results often surprise first-time users: Vitamin D and Exercise Score routinely appear at the top of the ranking, ahead of most drug doses. This is not a bug — it reflects the ODE model's validated representation of how patient biology shapes treatment response. If those parameters were entered as rough estimates, consider re-running with adjusted values to bracket the uncertainty.



Once you have reviewed the metrics and chart, click the green 'Tab 4 — Dose Optimisation (GA)' button in the completion banner to proceed. You can also click tab number 4 in the top bar, or select 'GA-Based Optimal Dose Plan' in the sidebar.

5. Tab 4 — GA-Based Optimal Dose Plan

Tab 3 showed you what happens with a fixed dose plan. Tab 4 asks a different question: what dose plan would produce the best possible outcome for this specific patient? It answers that question using a genetic algorithm (GA) — a search method inspired by biological evolution that explores thousands of dose combinations and progressively refines them toward a clinical optimum.

Click 'Start GA Optimisation'. The algorithm runs 10 generations of 8 candidate solutions, evaluating each candidate by running the full ODE simulation. Progress stages appear on screen as the algorithm works. Expect 30–90 seconds depending on how many drugs are active.

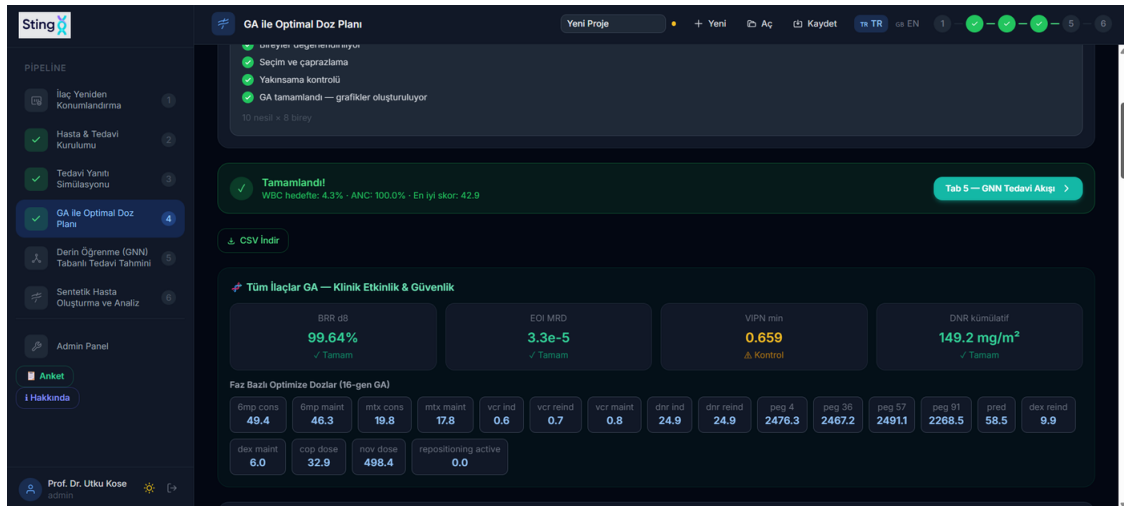


Figure 9. Tab 4 immediately after optimisation completes. The four metric cards each display the value achieved by the optimal plan, with a ✓ (target met) or ⚠ (monitor) status badge. The blue 'Tab 5 — GNN Treatment Flow' button appears in the completion banner.

5.1 Reading the Four Metric Cards

When the optimiser finishes, four cards appear. Each represents a clinical target the GA was asked to satisfy simultaneously:

Metric	Target	How to read the result
BRR d8	$\geq 80\%$	Day-8 blast reduction rate — how much of the leukaemic burden the steroid phase eliminated by day 8. In the example run, 99.64% means an extremely strong early steroid response was predicted. A ✓ badge confirms the target was met.
EOI MRD	< 0.01	End-of-induction minimal residual disease. The example value of 3.3×10^{-5} represents a deeply MRD-negative response — one of the strongest prognostic indicators in ALL. ✓ here means the model predicts MRD negativity.
VIPN min	≥ 0.78	The neuropathy safety threshold. If the GA could not keep VIPN above 0.78 while also meeting the other three targets, a ⚠ badge appears here. This usually means VCR dosing is pressing against the neuropathy boundary — the Dose Contribution chart (below) will confirm this.
Cumulative DNR	$< 300 \text{ mg/m}^2$	The cardiotoxicity ceiling. The GA treats this as a hard constraint; it will never propose a plan that exceeds 300 mg/m^2 even if doing so would improve other metrics.

5.2 Exploring the Optimal Dose Plan Chart

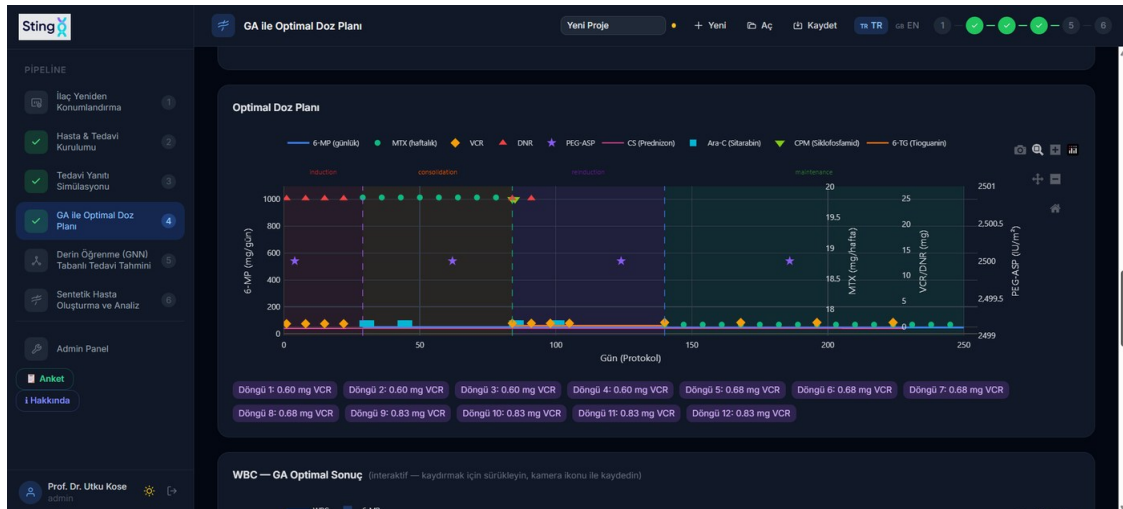


Figure 10. The Optimal Dose Plan chart. Each drug's administration events appear as coloured symbols along the time axis. The four coloured phase regions (induction · consolidation · reinduction · maintenance) are separated by dashed vertical lines. Below the chart, numbered boxes list the VCR dose for each administration cycle.

Below the metric cards, this chart maps out when and at what dose each drug should be given. The x-axis is treatment day (0–250); the y-axis scales independently per drug type — 6-MP and MTX share the left axis, VCR - DNR share the right axis, and PEG-ASP has its own far-right axis.

Read it like a clinical schedule: each data point is one administration event. A triangle at day 22 in the DNR row means one DNR dose is scheduled on day 22, at the dose value shown on the y-axis. The numbered boxes below the chart list the specific VCR dose for each of the 12 administration cycles — these values change between cycles because the GA found different optimal doses for different phases.

5.3 Dose Contribution Analysis — Understanding What the GA Decided

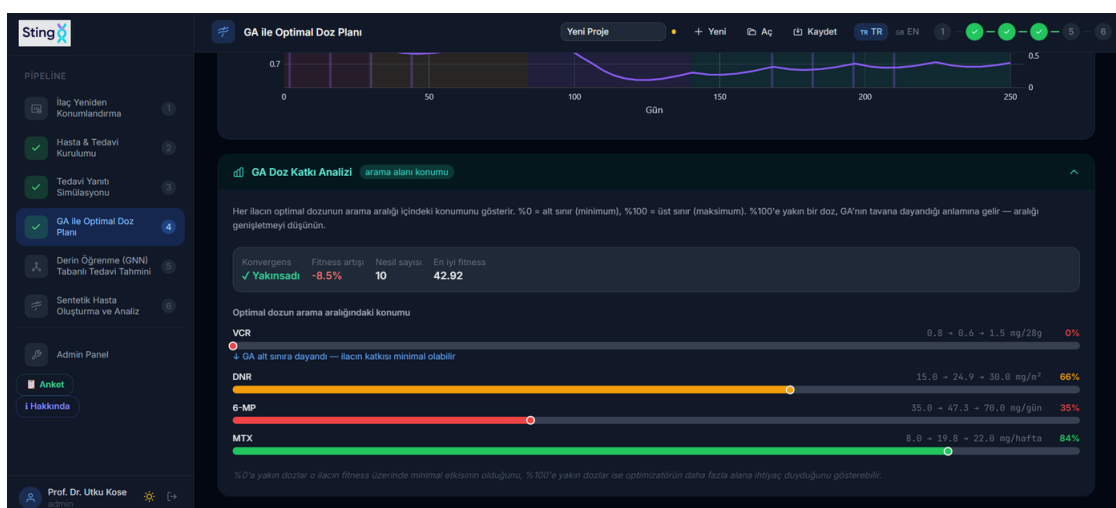


Figure 11. GA Dose Contribution Analysis. Each horizontal bar shows where the GA's chosen dose sits within the allowed search range — 0% is the minimum allowed, 100% is the maximum. MTX at 84% and DNR at 66% are near the top of their ranges. VCR at 0% is sitting right at its minimum.

This section is the most interpretively useful output in Tab 4. Each drug gets a labelled bar showing where the GA's chosen dose falls within its allowed search range — 0% means the GA picked the lowest allowed dose, 100% means it picked the highest. The numbers beside each bar make it concrete: for example, 'VCR: 0.8 → 0.6 → 1.5 mg/28g 0%' means the search range was 0.8 to 1.5 mg/28g, and the GA settled at 0.6 mg (effectively the lower bound).

- MTX at 84% — the GA pushed methotrexate close to its upper boundary. This suggests higher MTX is beneficial for this patient and the search range could potentially be widened if clinical safety allows.
- DNR at 66% — mid-to-high range. The GA is balancing efficacy against the 300 mg/m² cardiotoxicity ceiling; you can see the tension reflected in this position.
- 6-MP at 35% — mid-range, consistent with standard dosing. No strong signal to adjust the search boundaries.
- VCR at 0% — the GA hit the lower bound and stayed there. This is not an error; it is the GA telling you that reducing VCR dose improved the VIPN score without meaningfully sacrificing the other targets. If the ⚠ badge appeared on the VIPN card, this is why: even at the minimum, VCR's neuropathy effect was challenging to contain.



The GA result is automatically saved to the pool that Tab 5 reads. When you open Tab 5, you will see this run listed as a selectable record. Click it to load the optimised plan into the GNN and XAI analysis.

6. Tab 5 — Deep Learning (GNN) Treatment Prediction and XAI

Tab 5 is where the system explains its reasoning. The GNN (Graph Neural Network) digital twin acts as a fast approximator of the ODE: given a patient profile and dose plan, it predicts treatment trajectory outcomes in milliseconds instead of seconds. On top of that speed advantage, it connects to four XAI methods that let you ask specific questions about why the model predicts what it predicts.

The first thing you will see on Tab 5 is a list of GA optimisation records. Each row represents one past optimisation run. Click the row from your current session — it highlights in blue and a 'Selected for XAI' badge appears beside it. All four XAI analyses below will then operate on that record.

6.1 Choosing a Target Output

Before clicking any XAI method's Run button, select the clinical variable you want to explain. The eight pill-shaped buttons near the top of the XAI section represent the outputs the GNN predicts:

- WBC — white blood cell trajectory
- ANC — absolute neutrophil count
- Lt — leukaemic cell burden
- VIPN_N — neuropathy index
- PEG_A — asparaginase activity
- ASN — asparagine level
- cum_DNR_mgm2 — cumulative daunorubicin
- CCS — composite clinical score

Click the pill for the variable you are most interested in, then run whichever XAI method you want. You can change the target at any time and re-run any method independently — each method runs its own separate analysis.

6.2 SHAP — Feature Contributions

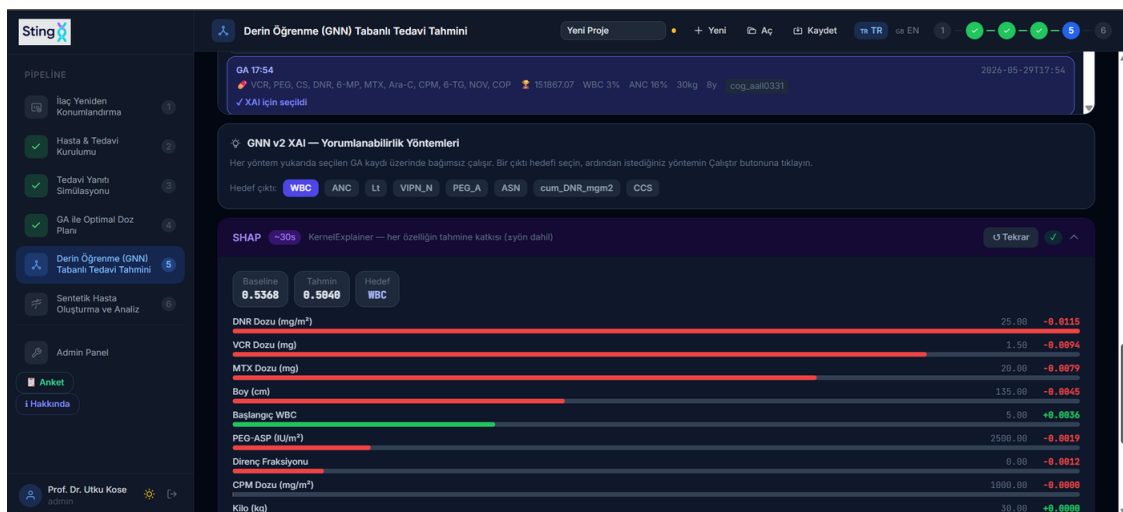


Figure 12. SHAP results with WBC as the target output. Each horizontal bar is one input feature. Red bars reduce the predicted WBC; green bars increase it. The length of the bar shows how strong the effect is, and the number on the right is the signed SHAP score.

Click 'Run' in the SHAP panel (the label shows ~30s). SHAP (SHapley Additive exPlanations) assigns each input feature a contribution score that quantifies how much that feature is pushing the prediction higher or lower, averaged consistently across all possible feature orderings.

Looking at the chart: every red bar is a feature that pulled the WBC prediction downward — higher doses of cytotoxic drugs suppress blood cell production, so DNR Dose (-0.0115), VCR Dose (-0.0094), and MTX Dose (-0.0079) appear as the three strongest negative contributors, exactly as clinical intuition would predict. The one green bar — Initial WBC ($+0.0036$) — makes sense too: a patient who starts with higher WBC has more headroom before the count drops into dangerous territory.

If a feature you expected to be important appears near the bottom of the list with a near-zero score, that is meaningful information: the model has learned that feature does not strongly influence the chosen output at the current parameter values.

6.3 Counterfactual — What Would Need to Change?

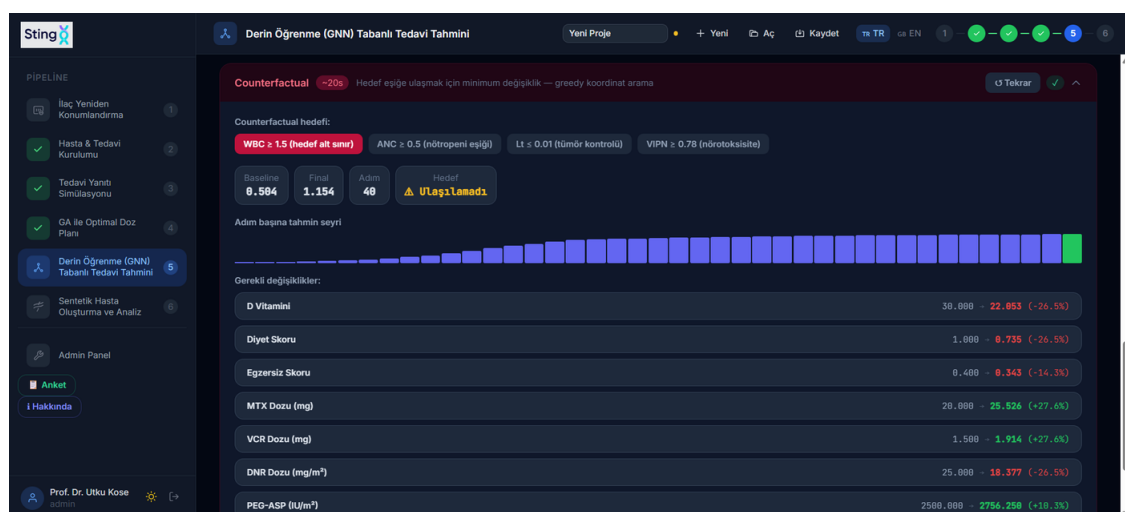


Figure 13. Counterfactual XAI panel. The red pill at the top shows the selected target ($WBC \geq 1.5$). The progress bar in the middle shows 40 search steps were taken. The 'Target not reached' status badge means the search found improvements but could not fully close the gap within bounds. The table below lists every parameter the search modified.

Click 'Run' in the Counterfactual panel (~20s). This method answers a directly actionable question: given the current predicted outcome, what is the minimum set of parameter changes that would shift the prediction to meet a chosen clinical threshold?

Select a target using the four buttons at the top of the panel:

- $WBC \geq 1.5$ — keep WBC above the haematological lower safety boundary
- $ANC \geq 0.5$ — avoid febrile neutropenia territory
- $Lt \leq 0.01$ — achieve tumour control
- $VIPN \geq 0.78$ — stay within neuropathy safety limits

The search begins at the current parameter values (Baseline) and iteratively nudges actionable features — dose values, lifestyle factors — looking for the shortest path to the target. After the search completes, the panel shows:

- Baseline value — where the prediction started
- Final value — the best value the search reached
- Steps taken — how many iteration steps the algorithm needed
- Target status — 'Reached ✓' or 'Not reached ⚠', depending on whether the threshold was crossed

The 'Required changes' table below lists every parameter the algorithm moved, with the original value on the left, the proposed value on the right, and the percentage change in brackets. Parameters the algorithm left unchanged are not listed. If the target was not reached, these are still the changes most likely to move the outcome in the right direction — they represent a ranked clinical action plan even when the full target remains out of reach.

6.4 GEMEX — Manifold-Based Explainability



Figure 14. GEMEX output. The Manifold κ and Arc Length values appear in the top two boxes. The geodesic arc-length profile (orange-red bar chart) shows how sharply the patient's feature vector bends away from the reference population along each step of the geodesic path. The ranked feature list below it shows GSF scores — Resistant Fraction is highest at +0.1629.

Click 'Run' in the GEMEX panel (~20s). GEMEX (Geodesic Entropic Manifold Explainability, v1.2.2) was developed within the STING project and takes a fundamentally different approach to explanation from SHAP. Where SHAP assumes linear additive contributions, GEMEX operates on the Riemannian manifold of the feature space — meaning it captures non-linear effects and interactions that SHAP can miss.

Three outputs appear after the calculation:

- Manifold κ (kappa) — a curvature value for the data manifold around this patient's position. Higher κ means the prediction landscape is more 'hilly' near this patient — small parameter changes produce larger-than-expected output swings. The example value of $\kappa = 0.5612$ indicates moderate curvature.
- Geodesic arc-length profile — the bar chart in the upper panel. Reading it left to right: each bar represents one step along the geodesic path from the reference population centre to this patient's position in feature space. Tall bars indicate where the path bends most sharply — where this patient is most 'unusual' relative to the training distribution.

Bars clustered at the start of the path mean the patient is quickly diverging from the reference population. The colour gradient from orange to deep red also signals increasing geodesic distance.

- GSF scores (feature list) — the ranked list of Geodesic Sensitivity Field scores below the bar chart. The highest-scoring feature (Resistant Fraction at +0.1629 in the example) is the one whose variation most strongly bends the geodesic path — meaning small changes in that feature produce the largest shifts in the prediction manifold. This is a different and complementary signal to SHAP: GEMEX identifies which features create prediction instability, not just which ones push the value up or down.

6.5 GNN Model Management



Figure 15. GNN v2 Training Panel. The GCNConv architecture diagram is displayed during training — two GCNConv layers, a dropout layer, and a linear output layer. The green terminal-style console at the bottom streams training loss values epoch by epoch.

Scroll to the bottom of Tab 5 to reach the model management section. The active model's filename and training date are shown at the top. Below that, three panels give you control over the model:

- Upload a Custom Model — click 'Upload Model (.pth)' to replace the active GNN with one you have trained externally. The format must be a PyTorch state dict saved with `torch.save()`. The system validates the file and activates the new model immediately.
- Restore Default Model — if a custom upload produces unexpected XAI results, click this button to revert to the project's validated pre-trained model (trained on 150 GA records, median $R^2 = 0.991$ across eight output targets).
- Train GNN v2 with Pool Data — the training panel below lets you retrain the GNN on the current GA pool. Set Epoch (default 150), Learning Rate (0.001), Hidden Layer size (256), and Dropout (0.2). Click 'Start Training'. The architecture diagram animates during training, and the console at the bottom streams epoch-by-epoch loss values in real time. Training typically completes in 2–5 minutes. When the loss stabilises and the console stops updating, training is done.



After training, download the new model using 'Download Model (.pth)' before navigating away. The file persists on the server but may be overwritten by a future training run if you do not save a copy.

7. Tab 6 — Synthetic Patient Generation and Analysis (GAN)

Tab 6 shifts the perspective from a single patient to an entire patient population. The CTGAN (Conditional Tabular GAN) generative model produces cohorts of statistically realistic synthetic ALL patients, each automatically assigned a 5-class risk stratification. Then — and this is the feature that makes Tab 6 unique — the repositioning scenario panel re-runs the ODE for the entire cohort with Copanlisib and Novobiocin added, giving you a cohort-level before-and-after view of the repositioning effect.

When Tab 6 opens, it is in 'Use Pre-trained Model' mode by default. Start here — the pre-trained model and its reference CSV are already validated and ready.

7.1 Generating a Cohort

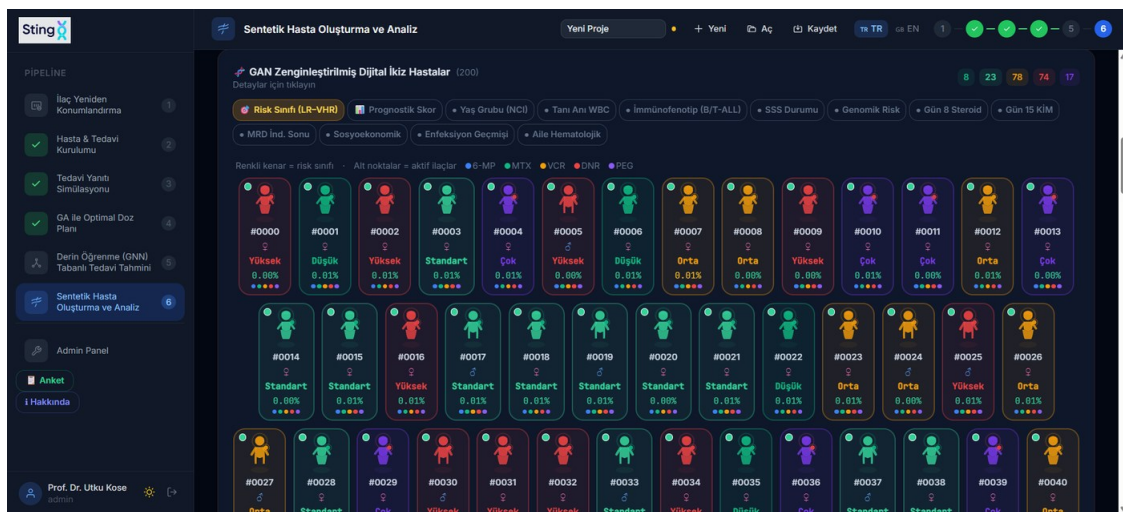


Figure 16. Tab 6 after generating 200 patients. The risk class distribution (8 LR · 23 SR · 78 IR · 74 HR · 17 VHR) appears in coloured numbers at the top-right of the grid. Each card shows a colour-coded silhouette, patient ID, risk class label, and end-of-induction MRD value.

Two controls appear at the top of the patient grid:

- Patient Count — type a number between 1 and 500. Start with 20–50 for a quick overview; use 200 for a population-level analysis. Larger cohorts take proportionally longer — 200 patients typically finishes in under 30 seconds.
- Seed — an integer that fixes the random number generator, making results reproducible. Leave it blank for a fresh random cohort each time, or enter any number to pin the generation. Useful when you want to compare the same cohort under different repositioning doses.

Click 'Generate Patients' (or '+ Add 10 Profiles'). A progress bar appears briefly, then the grid fills with patient cards. Each card has a coloured border that immediately tells you the risk class: green for LR, blue for SR, orange for IR, red for HR, purple for VHR.

7.2 Reading a Patient Card in Detail

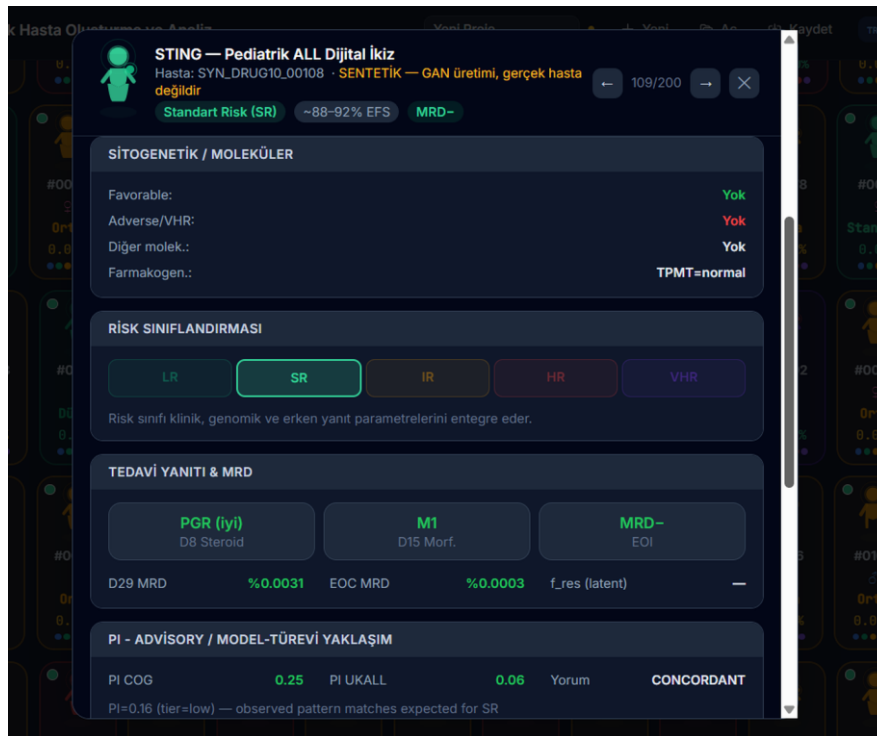


Figure 17. A patient detail modal (SR risk class, MRD-negative). The nine information sections are stacked vertically and can be scrolled. The ← and → arrows in the header navigate to adjacent patients without closing the modal.

Click any card to open the detail modal for that patient. This is the most information-dense view in the application — it presents everything the system knows about the patient across nine collapsible sections. You can navigate between patients with the ← and → arrows without closing the modal.

Walk through the sections from top to bottom:

- Identity and Anthropometry — age, sex, weight, height, BSA. Age, sex and ethnicity are generated by the GAN; weight and height are derived from age using paediatric growth reference ranges, and BSA is computed from them.
- Diagnosis — WBC at diagnosis, ALL subtype (B-ALL or T-ALL), CNS status, testicular involvement, extramedullary disease, and infection history.
- Cytogenetics / Molecular — favourable markers (ETV6-RUNX1, high hyperdiploidy), adverse/VHR markers (BCR-ABL1, hypodiploidy, KMT2A-R), and pharmacogenetic status (TPMT, NUDT15, anti-asparaginase antibody).
- Risk Classification — the five-class band with the active class highlighted in colour and a short explanation of the criteria that determined it. 'SR because: NCI-SR + PGR + M1 + EOI MRD <0.01' is the kind of plain-language reasoning shown here.
- Treatment Response and MRD — Day 8 steroid response badge (PGR = good / PPR = poor), Day 15 bone marrow morphology (M1 / M2 / M3), EOI MRD badge, D29 MRD percentage, and end-of-consolidation MRD. These values are calculated post-hoc from the GAN outputs, not predicted by the GAN itself.
- PI Advisory — Prognostic Index scores from both COG and UKALL frameworks, shown side by side, with a CONCORDANT or POTENTIAL_DISCORDANCE summary.

Concordant means both frameworks agree; discordance may be clinically significant and worth noting.

- Prognosis — 5-year EFS and OS ranges drawn from the assigned risk class, plus a relapse category label.
- Treatment Protocol — a compact four-phase dose table showing BSA-scaled doses for every active drug in each phase.
- Safety and DSS — cumulative DNR dose and minimum VIPN score, each with a colour-coded indicator showing how they relate to the safety thresholds.

7.3 The Population Summary Panel

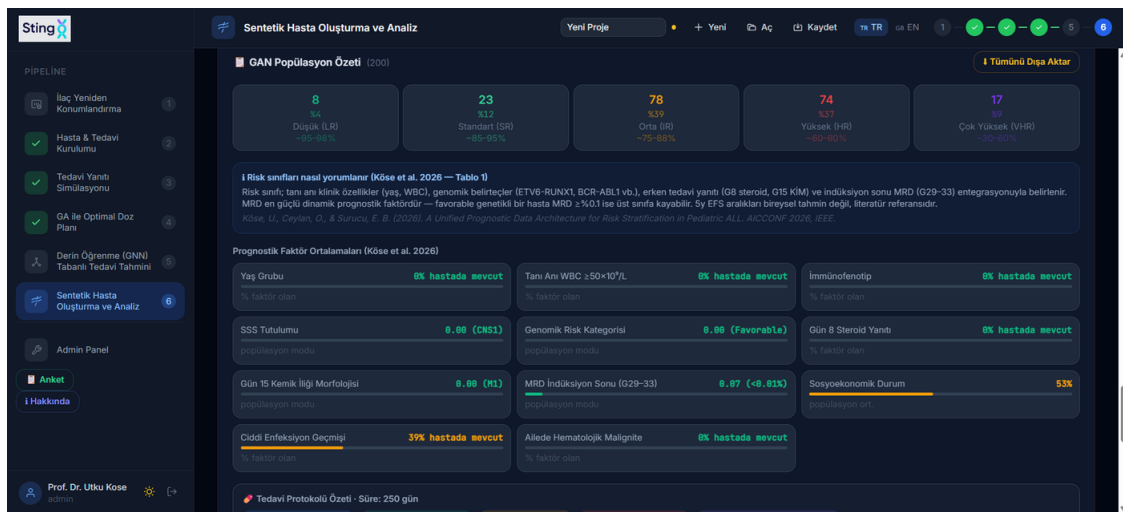


Figure 18. Population Summary panel. The top row shows risk class counts (8 LR · 23 SR · 78 IR · 74 HR · 17 VHR) and their 5-year EFS ranges. The grid below shows population-level averages for key prognostic factors — CNS involvement, MRD, genomic risk, and socioeconomic status.

Scroll above the patient grid to reach the Population Summary panel. This gives you the cohort-level view instead of the individual patient view.

The top row breaks down how many patients landed in each of the five risk classes, with the expected 5-year EFS range shown for each class. In a typical 200-patient run you will see something like 8 LR (4%), 23 SR (12%), 78 IR (39%), 74 HR (37%), and 17 VHR (9%). The heavy skew toward IR and HR reflects the ODE parameter ranges used during generation — the system conservatively models more challenging clinical presentations.

The prognostic factor grid below the counts shows population averages for diagnostically important variables: CNS involvement, genomic risk category, MRD at end of induction, Day 8 steroid response, and socioeconomic status. Compare these averages to published ALL epidemiology to assess how representative your generated cohort is.

The 'Export All' button in the top-right corner of the panel downloads the entire cohort as a CSV file — one row per patient, one column per clinical attribute. This is the primary way to take cohort data out of the system for external analysis.

7.4 The Repositioning Scenario — COP + NOV at Cohort Level

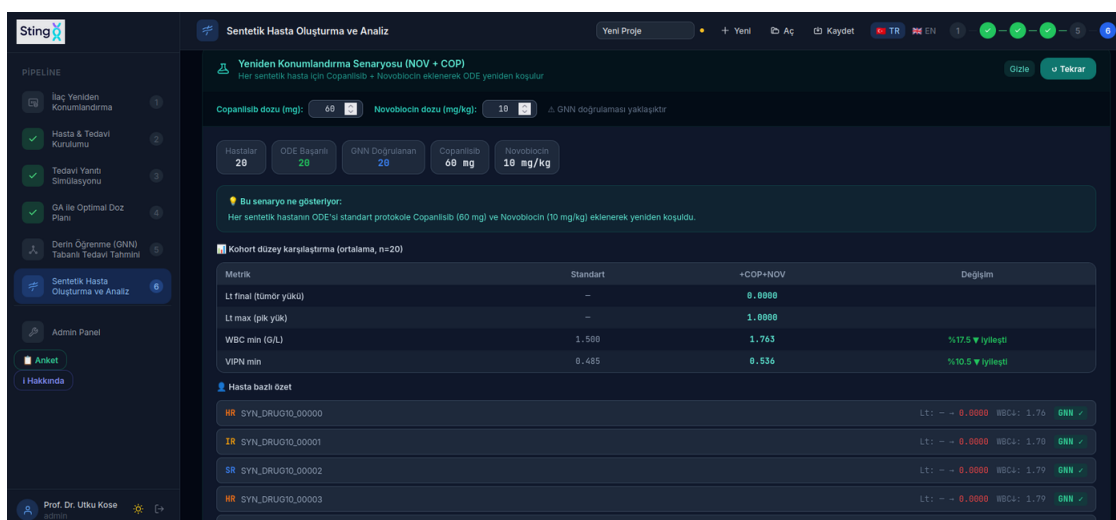


Figure 19. Repositioning Scenario panel after running a 20-patient cohort. The dose controls (Copanlisib 60 mg, Novobiocin 10 mg/kg) appear at the top. The comparison table in the middle shows cohort-mean values for Lt final, WBC min, and VIPN min under Standard vs +COP+NOV protocols. The per-patient list at the bottom shows each patient's risk class, Lt trajectory, minimum WBC, and GNN validation badge.

Scroll to the bottom of Tab 6 to reach the 'Repositioning Scenario (NOV + COP)' panel. This is the feature that closes the loop between the drug discovery work in Tab 1 and the patient population in Tab 6. It takes every patient in the current cohort and re-runs their ODE simulation with Copanlisib and Novobiocin added to the standard protocol — then presents a side-by-side before-and-after comparison.

The reason this is architecturally significant: the GAN model was trained on data that does not include COP or NOV. Their effect is introduced entirely through the ODE layer during post-hoc calculation, which means you are seeing a mechanistically grounded prediction of what these repositioned drugs would do — not an interpolation from training data.

To run the scenario:

1

Set doses

The Copanlisib dose field defaults to 60 mg; the Novobiocin field defaults to 10 mg/kg. Adjust these if you want to explore a different dose level. Clicking the up/down arrows changes the value in 5 mg / 1 mg/kg increments.

2

Click 'Re-run Scenario'

A spinner appears while ODE simulations complete for each patient in the cohort. For 20 patients this takes about 15–30 seconds; for 200 patients, allow up to 3 minutes.

3

Read the comparison table

The table shows cohort-mean values for four metrics: Lt final (leukaemic burden at day 250), Lt max (peak burden), WBC min, and VIPN min. Each metric shows the value under the standard protocol and the value with COP+NOV added, plus the percentage change and a directional label. A change from 1.500 to 1.763 for WBC min, labelled '+17.5% ▼ improved',

means the ODE predicts less WBC suppression when COP+NOV are included — a clinically important finding.

4

Check the per-patient list

Below the table, every cohort patient appears in a scrollable list. Each row shows the patient ID with their risk class badge, the Lt trajectory (standard value → COP+NOV value), the minimum WBC, and a GNN validation badge. 'GNN ✓' means the updated trajectory falls within the region of the feature manifold the GNN has reliable knowledge of; 'GNN ~' means it is extrapolating.

In the example run shown, Lt dropped to 0.0000 for all 20 patients — the model predicts complete leukaemic burden elimination — while WBC minimum improved by 17.5% and VIPN minimum improved by 10.5%. All 20 patients received GNN ✓ badges, indicating the trajectories are within the validated prediction range. These results constitute an in-silico hypothesis for laboratory follow-up, not a clinical recommendation.

Note

The GNN digital twin was not retrained on Copanlisib or Novobiocin data. Its GNN ✓ badge means the trajectory shape is geometrically consistent with the learned manifold — not that the GNN has specific pharmacological knowledge of these drugs. All repositioning scenario outputs should be treated as exploratory signals requiring experimental validation.

7.5 Training a New GAN Model

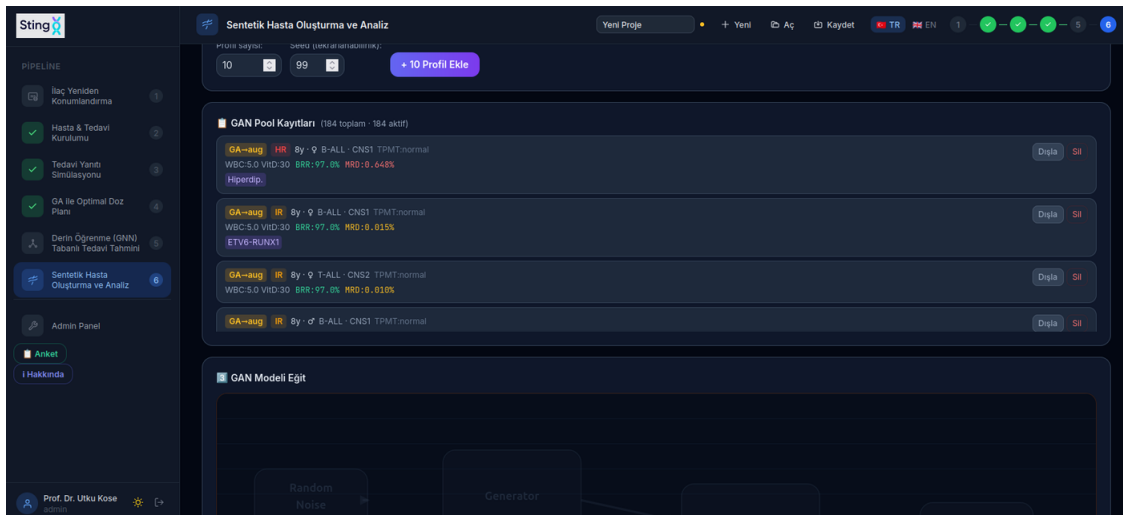


Figure 20. GAN Pool Records panel showing 184 active training records. Each row shows the source tag (GA→aug), risk class badge, key clinical values (WBC, Vitamin D, BRR, MRD), and any genetic markers. The Exclude and Delete buttons on the right let you curate the pool before training.

Switch to 'Train New Model' mode (the tab below the mode heading at the top of the page) to open the GAN training workflow. It has three stages:

1

Convert GA Pool → GAN Pool

Click the 'Convert GA Pool' button. This pulls records from the GA optimisation pool and processes them through the clinical_enrichment and risk_covariate_augmentation pipelines to produce GAN-ready training records. The conversion completes in seconds and populates the GAN Pool Records list below.

2

Curate the pool

Scroll through the 184 (or however many you have) records. Each row shows the source, risk class, and key clinical indicators. If any record looks like an outlier or a duplicate — unusually extreme WBC, implausible MRD, etc. — click 'Exclude' to remove it from training. Excluded records are dimmed and can be re-included by clicking 'Include' on the same row.

3

Train and download

Set the Epoch count (default 300) and confirm the Batch Size is a multiple of 10 (required by CTGAN's PAC parameter). Click 'Start Training'. A real-time loss chart shows Generator and Discriminator training curves as they evolve. When the curves stabilise and the console shows the final epoch, training is done. Click 'Download Model (.pkl)' and 'Download Reference CSV' to save both output files.



Note

After downloading the PKL and CSV files, switch back to 'Use Pre-trained Model' mode and upload both files using the model panel in the upper-right of the page. If you skip this upload step, the system continues generating patients from the old model even though training completed successfully.

8. Admin Panel

The Admin Panel link appears in the bottom-left sidebar and is visible only to users with the 'admin' role. Clinician accounts do not see it.

Sub-tab	What you can do here
User List	View all registered accounts with email, full name, and role. Each row has four action buttons: Edit (change name, email, or role), Reset PW (send a new temporary password), Deactivate / Activate (suspend or restore access without deleting the account), and Clear Logs (remove all activity records for that user without affecting their account).
Add User	Create a new account by entering username, email address, full name, initial password, and role (admin or clinician). The account becomes active immediately.
Activity Logs	A chronological record of every system action — which user, which tab, which operation, and the outcome. Use the filter controls to search by user or date range. 'Export to Excel' downloads the full log as an .xlsx file.
Surveys	View TÜBİTAK research survey responses submitted by users. Completion counts are shown per survey type (General Evaluation and ALL-specific). 'Export to Excel' downloads individual responses for analysis.

9. Common Questions

Situation	What to do
I changed a parameter in Tab 2 after running Tabs 3 and 4.	Re-run the ODE simulation in Tab 3, then re-run GA optimisation in Tab 4, then re-select your GA record in Tab 5. Tab 6 generates synthetic patients independently and does not need to be re-run unless you want a different cohort.
The XAI panels show error or returns a 502 response.	The analysis timed out on the server. Wait 60 seconds and click 'Run' again. If the problem repeats, try during off-peak hours — XAI panels have compute-intensive operations.
The GA result shows ⚠ on the VIPN min card.	The optimiser could not satisfy all four clinical targets simultaneously. Look at the Dose Contribution chart — if VCR is at 0%, it is already at the lower search bound. Consider widening the lower bound or accepting a slightly reduced BRR d8 target. You can also re-run the GA after reducing the VCR dose search range minimum.
A GAN patient has a risk class that seems inconsistent with its clinical data.	Open the patient detail card and expand the Risk Classification section — it explains exactly which criteria determined the class. The classification is rule-based; borderline cases may fall into an adjacent class depending on which threshold the specific GAN-generated values crossed.
I want to reproduce a specific cohort from a previous session.	Re-enter the same patient count and seed value that you used in the previous session and click 'Generate Patients'. The same seed always produces the same cohort from the same model version.
My session expired and I lost unsaved results.	Sessions time out after 8 hours of inactivity. Use the 'Save' button in the top bar after completing each tab to avoid data loss. Saved sessions can be reopened with 'Open' at any future login.